

Blockchain Health Insurance Project Roadmap

Project Objective

- 1 Build a decentralized platform to manage medical records and insurance claims.
- 2 Automate reimbursement using smart contracts.
- 3 Ensure transparency, security, and traceability.

MVP Features

- 1 User registration and role management
- 2 Insurance policy creation and assignment
- 3 Medical record submission with document hash
- 4 Claim submission and automatic reimbursement calculation
- 5 Approval/rejection workflow
- 6 History tracking

Architecture Overview

- 1 Smart Contracts (Solidity)
- 2 Frontend (React + Ethers.js)
- 3 Backend (Node.js or FastAPI)
- 4 Storage: IPFS for documents, optional DB for metadata

Roles Distribution

- 1 Member 1: Smart Contracts – Solidity, Hardhat, business logic, testing
- 2 Member 2: Frontend – React UI, wallet integration, dashboards
- 3 Member 3: Backend – IPFS integration, API, system integration

Development Phases

- 1 Phase 1: Requirements and architecture design
- 2 Phase 2: Smart contracts implementation
- 3 Phase 3: Backend and IPFS setup
- 4 Phase 4: Frontend development
- 5 Phase 5: Integration and testing
- 6 Phase 6: Final demo preparation

Workflow Example

- 1 Admin registers actors
- 2 Insurer creates policy
- 3 Patient subscribes
- 4 Doctor submits medical record

- 5 Claim is generated
- 6 Smart contract computes reimbursement
- 7 Insurer approves/rejects
- 8 Patient views history